



INTERNSHIP REPORT

Under NEP 2020 (National Education Policy 2020)

Submitted To:

**ANAND MERCANTILE COLLEGE OF SCIENCE,
MANAGEMENT AND COMPUTER TECHNOLOGY,
ANAND**

(Affiliated to Sardar Patel University, Vallabh Vidyanagar)



Internship conducted at:

Information Technology Canter

Duration of Internship:

From 03/11/2025 To 26/11/2025

Programme & Semester:

BCA (Semester – VI)

Submitted By:

Mohamad Hasnen Mohamad Rashad Shaikh

Roll Number:

3052

Under the Guidance of:

(Faculty Mentor / Guide Name)

Academic Year:

2025 – 2026

DECLARATION

I, the undersigned, **Mohamad Hasnen Mohamad Rashad Shaikh**, student of **BCA SEM – 6** at **Anand Mercantile College of Science, Management and Computer Technology, Anand**, hereby declare that the project titled “**House Price Prediction**” is the result of my own work carried out during my **Internship under the New Education Policy (NEP) – 2020**.

I further declare that this project is an original work and has not been submitted, in whole or in part, for any other degree, diploma, or certificate at any other institution. All sources of information and references used in this project have been duly acknowledged.

I also affirm that the project work has been completed under the guidance and supervision of _____ (**Name of Guide/Mentor**) at _____ (**Name of Organization/Company**) as part of my internship program.

I take full responsibility for the authenticity of the contents and data presented in this report.

Place: Anand

Date: _____

Signature of Student

Name: _____

Class/Semester: _____

Roll No.: _____

Anand Mercantile College of Science, Management and Computer Technology, Anand

INTERNSHIP COMPLETION CERTIFICATE

This is to certify that **Mr. Mohamad Hasnen Mohamad Rashad Shaikh**, Student of **BCA programme semester VI**, has successfully completed a **21 Days (120 hours)** Internship Program under the **New Education Policy (NEP) 2020** at **Information Technology Center**, from **03/11/2025 to 26/11/2025**.

During the internship, the student showed professionalism, a willingness to learn, and the ability to handle assigned tasks responsibly. They demonstrated good communication skills, applied theoretical knowledge effectively, and worked well in teams while adapting to workplace requirements. The student has prepared a **Internship Report** based on the learning outcomes and experience acquired during the internship period, as a part of the academic requirement for the degree program.

We appreciate the student's sincere efforts and wish him/her success in all future endeavours.

Date:

Place: Anand

(Internship Supervisor / Faculty Guide)

(Head / Principal)

ACKNOWLEDGEMENT

I express my sincere gratitude to **Anand Mercantile College of Science, Management & Computer Technology and Sardar Patel University**, for providing me the opportunity to undergo an internship under the **New Education Policy (NEP) 2020**. This internship has given me valuable practical exposure and a chance to apply my academic knowledge in a real organizational environment.

I am deeply thankful to my **Principal, Dr. Rupal D. Bhatt Ma'am**, for her constant encouragement and valuable guidance throughout the internship period. Her motivation has been a great source of inspiration in the successful completion of this project.

I would like to express my heartfelt thanks to my **faculty mentor/ guide**, _____, for her/his continuous guidance, valuable suggestions, and support during the preparation of this project report. I truly appreciate their dedication and the time they invested in monitoring my progress and ensuring the successful completion of my internship.

I am also thankful to the management and staff of **Information Technology Centre** for providing me the opportunity to work as an intern and for their cooperation and assistance throughout my internship period. Their insights and encouragement helped me complete my tasks with confidence and clarity.

My sincere gratitude goes to my **parents and family members** for their moral support, patience, and encouragement, which have been my greatest strength throughout this journey.

Lastly, I extend my appreciation to all my **friends, classmates, and well-wishers** who directly or indirectly helped me in the successful completion of this internship project.

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1. Introduction

1.1 About the Internship

The internship was designed to provide practical exposure to Python programming, data handling, and basic machine learning concepts. It helped me understand how AI and ML are applied in real-world scenarios. It helped bridge the gap between academic knowledge and industry practices through hands-on tasks, mentorship, and exposure to organizational workflows.

1.2 Organization Profile

The internship was conducted under the guidance of **Information Technology Centre (ITC)**, which provides training in emerging technologies like Artificial Intelligence, Machine Learning, Data Science, and Python programming. The organization focuses on skill development and industry-oriented learning.

1.3 Duration

The duration of the internship was **21 days** and **120 hours**.

Each day included learning sessions, practical assignments, and project-related activities.

- **Total Duration:** 21 Days
- **Total Hours :** 120 hours
- **Daily Working Hours:** 6 Hours
- **Mode:** Practical and Theory Based Learning

The internship was well-planned and divided into different phases, starting from basic programming to advanced machine learning concepts.

1.4 Technology Used

The following technologies and tools were used during the internship and project development:

- **Python:** Used as the main programming language
- **NumPy:** Used for numerical computations
- **Pandas:** Used for data manipulation and analysis
- **Matplotlib & Seaborn:** Used for data visualization
- **Scikit-learn:** Used for implementing machine learning models
- **Streamlit:** Used to create a simple web application for house price prediction
- **Jupyter Notebook:** Used for coding and testing models

These tools helped in understanding data processing, model training, evaluation, and deployment.

1.5 Role & Responsibilities

During the internship, my roles and responsibilities included:

- Learning Python programming fundamentals
- Practicing control statements and data structures
- Working with datasets using Pandas and NumPy
- Understanding machine learning algorithms
- Developing a mini project on House Price Prediction
- Creating a simple web interface using Streamlit
- Maintaining daily work logs

I actively participated in all internship activities and completed the assigned tasks within the given time.

2. Objectives of the Internship

2.1 Purpose of Internship

The purpose of this internship was to bridge the gap between theoretical learning and practical implementation. In classroom learning, students mainly focus on theory, but through this internship, I got the opportunity to apply my knowledge practically.

The internship helped me understand:

- How real-world data is collected and processed
- How machine learning models are trained and tested
- How predictions are made using data
- How a simple web application is developed using Streamlit

Another important purpose of the internship was to build confidence in programming and improve my understanding of Python and machine learning concepts. It also helped me learn time management, discipline, and teamwork. Overall, the internship was helpful in developing both technical and professional skills.

2.2 Technical Objectives

The technical objectives of this internship were focused on building strong fundamentals and practical experience in programming and machine learning.

The main technical objectives were:

- To learn Python programming from basic to intermediate level
- To understand control statements, loops, and functions
- To work with Python data structures such as lists, tuples, dictionaries, and sets
- To learn data handling using Pandas and NumPy
- To understand data visualization using Matplotlib and Seaborn
- To learn basic concepts of Machine Learning algorithms
- To implement a House Price Prediction model using machine learning
- To create a Streamlit web application for predicting house prices
- To understand model evaluation and performance analysis

These objectives helped me gain hands-on experience and practical exposure to machine learning tools and techniques.

2.3 Skills to be Developed

This internship was very helpful in developing both technical and non-technical skills. The skills gained during the internship are listed below:

Technical Skills:

- Python programming

- Data analysis and data preprocessing
- Machine learning model development
- Working with real-world datasets
- Data visualization
- Model testing and prediction
- Streamlit web application development

Non-Technical Skills:

- Problem-solving skills
- Analytical thinking
- Time management
- Communication skills
- Teamwork
- Self-learning ability

These skills will be very useful for my future academic projects and professional career in the IT field.

3. Learning Experience

3.1 Introduction to Python

Python is a high-level, interpreted programming language that is widely used in data science, machine learning, and artificial intelligence. During the internship, I learned the basics of Python programming from scratch.

I understood the importance of Python because of its simple syntax and readability. Topics such as variables, data types, operators, and input-output operations were explained clearly. Practical examples were given to understand how Python programs work.

Python helped me build a strong programming foundation, which was very useful for learning machine learning concepts later.

3.2 Control Statement

Control statements are used to control the flow of execution in a program. During the internship, I learned different types of control statements in Python.

The following control statements were covered:

- Conditional Statements: **if, if-else, elif**
- Looping Statements: **for loop and while loop**
- Loop Control Statements: **break, continue, pass**

By using control statements, I learned how to make decisions in a program and how to repeat tasks efficiently. Practical programs were written to understand conditions and loops properly. This topic helped me improve my logical thinking and problem-solving skills.

3.3 Functions & Modules

A **function** is a block of code that performs a specific, named task and only runs when it is called. Functions are used to divide a program into smaller and reusable parts. During the internship, I learned how to define and use functions in Python.

I learned:

- Creating user-defined functions
- Function parameters and return values
- Advantages of using functions

Modules are used to organize code and reuse existing functionality. I also learned how to import and use built-in Python modules.

This topic helped me write clean, structured, and reusable code, which is very important in real-world projects.

3.4 Python Library

Python libraries play a major role in data analysis and machine learning. During the internship, I learned about various important libraries:

- **NumPy:** Used for numerical operations and working with arrays
- **Pandas:** Used for data manipulation, data cleaning, and data analysis
- **Matplotlib:** Used for basic data visualization
- **Seaborn:** Used for advanced and attractive data visualization
- **Scikit-learn:** Used for implementing machine learning algorithms

Using these libraries, I learned how to load datasets, clean data, analyze patterns, and visualize results. These libraries helped me understand data better and prepare it for machine learning models.

3.5 Introduction to Machine Learning

Machine Learning is a part of Artificial Intelligence that allows machines to learn from data and make predictions without being explicitly programmed. During the internship, I learned the basic concepts of machine learning.

I understood the difference between:

- Supervised Learning
- Unsupervised Learning

I also learned about training data, testing data, features, labels, and model evaluation. Real-life examples were used to explain how machine learning works. This topic gave me a clear idea of how machine learning models are used in practical applications.

3.6 Machine Learning Algorithms Learn

During the internship, I learned various machine learning algorithms and their applications.

- **Linear Regression:** Used for predicting continuous values such as house prices
- **Logistic Regression:** Used for classification problems
- **K-Nearest Neighbors (KNN):** Used for classification and regression based on similarity
- **Decision Tree:** Used for decision-making and classification

Each algorithm was explained with examples, and practical implementation was done using Python and Scikit-learn. These algorithms helped me understand how predictions are made using data.

3.7 Mini Project Work

As part of the internship, I worked on a mini project titled “House Price Prediction Using Machine Learning”.

The objective of this project was to predict house prices based on various features such as area, number of bedrooms, location, and other factors.

Project Activities Included:

- Data collection from Kaggle
- Data preprocessing and cleaning
- Feature selection
- Model training using Linear Regression
- Model testing and evaluation
- Prediction of house prices
- Development of a simple web application using Streamlit

Streamlit was used to create a user-friendly interface where users can enter house details and get predicted prices. This project helped me understand the complete machine learning workflow from data collection to deployment.

3.5 Other Activities / Soft Skill

Apart from technical learning, the internship also helped me develop soft skills.

Some important soft skills developed during the internship are:

- Communication skills
- Teamwork
- Time management
- Presentation skills
- Self-learning ability

Daily discussions and interactions helped improve my confidence and professional behaviour. These skills are very important for future career growth.

4. Learning Outcome

4.1 Technical Skills Achieved

During the internship, I developed strong technical skills related to programming and machine learning. The following technical skills were achieved:

- Good understanding of Python programming fundamentals

- Knowledge of control statements, functions, and modules
- Experience in using Python libraries such as NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn
- Understanding of machine learning concepts and algorithms
- Ability to preprocess and analyze datasets
- Experience in building and testing machine learning models
- Knowledge of Linear Regression for prediction problems
- Basic understanding of Streamlit for web application development

These technical skills helped me understand the complete process of machine learning model development.

4.2 Practical Skill Gained

Apart from technical knowledge, this internship helped me gain practical and hands-on experience. The following practical skills were gained during the internship:

- Working with real-world datasets
- Data cleaning and preprocessing techniques
- Model training, testing, and evaluation
- Making predictions using machine learning models
- Debugging and error handling in Python programs
- Developing a simple and interactive web interface using Streamlit
- Managing project files and documentation

These practical skills improved my confidence and helped me understand how real-time projects are developed.

4.3 Overall Professional Development

The internship also contributed to my overall professional development. I learned how to work in a structured environment and follow project guidelines.

The following professional skills were developed:

- Time management
- Problem-solving ability
- Analytical thinking
- Communication skills
- Team coordination
- Self-discipline and responsibility

This internship prepared me for future academic projects and professional opportunities. It gave me a clear direction towards learning advanced technologies in the field of Artificial Intelligence and Machine Learning.

5. Conclusion / summary

5.1 Summary of Internship

The 21 days college internship in the field of Artificial Intelligence and Machine Learning was very informative and beneficial. During this internship, I learned Python programming, data analysis, and machine learning concepts.

The internship started with basic Python concepts and gradually moved towards advanced topics like machine learning algorithms and project development. I worked on a mini project titled “House Price Prediction Using Machine Learning”, which helped me understand the complete machine learning workflow.

This internship helped me gain hands-on experience, improve my problem-solving skills, and build confidence in working with real-world data. Overall, the internship was a great learning opportunity.

5.2 Challenges & Solutions

During the internship, I faced some challenges while learning new concepts and working on the project. However, with proper guidance and practice, these challenges were resolved.

Challenges Faced:

- Understanding machine learning concepts for the first time
- Working with large datasets
- Data cleaning and preprocessing
- Debugging errors in Python programs
- Selecting the correct machine learning algorithm

Solutions:

- Regular practice and revision of concepts
- Guidance from mentors and trainers
- Using online resources and documentation
- Step-by-step debugging of code
- Testing different algorithms to improve accuracy

These challenges helped me improve my learning ability and technical confidence.

5.3 Future Scope

The House Price Prediction project has a wide future scope and can be further enhanced in many ways.

Future improvements include:

- Using advanced machine learning algorithms
- Improving model accuracy with more data
- Adding more features like location and market trends
- Deploying the project on cloud platforms
- Enhancing the Streamlit web application with more features

This project provides a strong foundation for learning advanced machine learning and real-world application development. It will be useful for future academic projects and professional work.